

Azmani Sultana
Id 22201949
CSE471 Section 09
Assignment 03

① In the scenario, object database is most suitable, as it is designed to store complex objects while preserving inheritance, object identity and relationships with seamless object-oriented integration.

It is the best choice, because

- i. supports complex objects and relationships: It stores complex, interrelated objects while preserving relationships, inheritance and object identity.
- ii. Seamless integration with object-oriented programming: Object can be stored and retrieved directly without converting them into relationship tables, eliminating the object relational mismatch.
- iii. Improves performance and maintainability: Tight coupling between the application and the database reduces conversion overhead, making the system faster, more efficient and easier to maintain.

② a) ID: 22201949

Fields	Average size (characters)
Order number	$2 + 5 = 7$
Date	$2 + 5 = 7$
Customer ID	$2 + 5 = 7$
Last name	$0 + 5 = 5$
First name	$1 + 5 = 6$
State	$9 + 5 = 14$
Amount	$4 + 5 = 9$
Tax rate	$9 + 5 = 14$

b)

$$\begin{aligned}\text{Record size} &= 7+7+7+5+6+14+9+14 \\ &= 69 \text{ characters}\end{aligned}$$

$$\begin{aligned}\text{e) Total record size} &= \text{Record size} + (\text{Record size} \times \text{overhead}) \\ &= 69 + (69 \times 0.35) \\ &= 93.15\end{aligned}$$

d)

$$\begin{aligned}\text{Initial table size} &= \text{Summation of all the digits of ID} \\ &\quad \times 1200 \\ &= (2+2+2+0+1+9+4+9) \times 1200 \\ &= 29 \times 1200 \\ &= 34800\end{aligned}$$

$$\begin{aligned}\text{Initial table volume} &= \text{Initial table size} \times \text{total record size} \\ &= 34800 \times 93.15 \\ &= 3241620\end{aligned}$$

e) 1095 days = 3 years

Quarterly growth rate = 1500

$$\therefore \text{Growth rate} = 1500 \times (3 \times 4) = 18000$$

$$\begin{aligned}\text{Table volume} &= (\text{Initial table volume} + (\text{Growth rate} \\ &\quad \times \text{total record size} \times \text{time})) \\ &= 3241620 + (1500 \times 3 \times 4 \times 93.15) \\ &= 4918320\end{aligned}$$

③

i) Functional cohesion: Every part contributes to one task.

Example:

```
def calculate-tax(income):  
    tax = income * 0.15  
    return tax
```

ii) Sequential cohesion: Output of one step is input to the next.

```
def process-data(data):  
    cleaned-data = clean-data(data)  
    result = analyze-data(cleaned-data)  
    return result
```

iii) Temporal cohesion: Tasks executed at the same time.

Example:

```
def initialize-system():  
    load-configuration()  
    create-log-file()  
    connect-database()
```

iv) Logical cohesion: Tasks grouped by category but selected via flags.

```
def input-handler(type):  
    if type == "key board":  
        get-keyboard-input()  
    elif type == "file":  
        read-file-input()
```

v) coincidental cohesion: Random tasks grouped together.

def random-operation():

calculate-salary()

print-report()

validate-password()

④

